

Crime Pattern Detection Utilizing Power BI Visualizations on the Microsoft Fabric Data Platform With the Public data.police.uk Dataset

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Abstract—This paper presents a big data-driven platform for crime pattern detection using Power BI visualizations on Microsoft Fabric, built upon the public data.police.uk dataset. Law enforcement agencies require scalable analytics to extract actionable insights from large, complex datasets. We implemented a Data Lakehouse architecture to process around 8,500 crime data files in CSV format from multiple regions, with Python-based metadata cataloging for structured access to crime outcomes, stop-and-search records, and street-level incidents. Dataflows and Notebooks in Fabric addressed regional inconsistencies and enabled efficient data transformation. Power BI reports provided intuitive and interactive visualizations for exploring geographic and temporal crime trends. Performance testing demonstrated up to 40% faster query response times compared to traditional warehouses, and regional crime analysis that previously took days was completed within hours. The results indicated that the platform scaled efficiently while maintaining stable performance under growing data volumes. Our approach demonstrates how unified analytics and visualization environments can democratize access to crime data insights, supporting evidence-based policing and public safety decision-making.

Keywords—Data Platform, Microsoft Fabric, data.police.uk, Crime Analytics, Data Lakehouse Architecture

I. INTRODUCTION

Law enforcement organizations in the new AI era increasingly rely on advanced analytical approaches to derive insights from large and complicated crime databases. Traditional processing approaches frequently struggle with the quantity, complexity, and heterogeneity of modern police data, as well as the need for an appropriate Extract, Transform, Load (ETL) process. Microsoft Fabric [1] provides a unified analytics environment that addresses these difficulties by combining

information management, processing, and visualization into a single platform. In this work, we use Microsoft Fabric to create a comprehensive data platform for analyzing the public data.police.uk dataset, which includes crime statistics from England, Wales, and Northern Ireland.

Building robust analytical systems requires careful planning, design, and management [2], [3]. Effective platforms must support multidimensional data flows and ensure scalability through all stages of the data lifecycle [4], [5]. Integrating diverse structured and unstructured sources is critical for meaningful analysis, especially when dealing with data from multiple police regions, each with specific reporting standards and formats.

Organizations face ongoing challenges in optimizing performance, operational efficiency, and user experience when selecting Software-as-a-service (SaaS) platforms for data analysis. Reliable solutions demand systematic pipeline management, from data ingestion to integration, and must efficiently extract value from databases, APIs, and flat files. Careful evaluation of existing platforms is essential, considering both opportunities for KPI development and potential limitations.

Our research demonstrates how Microsoft Fabric processes crime data at scale. We implement a data lakehouse architecture to manage over 8,500 distinct CSV files from multiple police regions, addressing the challenges of data standardization and integration. Python-based metadata cataloging organizes crime outcomes, stop-and-search records, and street-level incidents, enabling both strategic overviews and detailed incident analysis within a single environment. Dataflows and

Notebooks are used to automate data transformation and quality checks, ensuring consistency and reliability across the dataset.

The use of Power BI [6] is crucial in this research while developing interactive visualizations as part of our research process. Using Power BI visualizations, stakeholders and government officials can quickly investigate geographic and chronological crime patterns, find hotspots, and drill into incident-level factors contributing to crime events. This capability promotes evidence-based choice-making and increases law enforcement agencies' ability to respond to emerging crime location patterns shown on the map.

The main contributions of our research work include:

- (1) A scalable data architecture using Microsoft Fabric's Lakehouse framework for effective storage, management, and scalability of the big data data.police.uk dataset.
- (2) Interactive visualization capabilities through Power BI integration, allowing stakeholders to gain actionable insights into crime trends and patterns using visualizations and other advanced analytics.
- (3) A metadata management system that enhances data discovery, access, and organization, streamlining the analytical workflow.
- (4) Evidence of how unified analytics platforms can transform public safety data into actionable intelligence, supporting both operational and strategic objectives.

The structure of this paper is as follows. Section I provides the research background, objectives, and main contributions, laying the groundwork for the study. The paper outlines the issues in crime data analytics and the motivation for adopting Microsoft Fabric. Section II describes the methodology, including research goals, data-gathering strategy, technological implementation using the data.police.uk dataset, and technical procedures followed to develop the platform. Section III evaluates the literature on comprehensive analytics platforms, Lakehouse design principles, and law enforcement data analysis. We highlight both established procedures and current gaps that our work addresses. Section IV discusses the implementation outcomes, specifically metadata cataloging. While integrating Microsoft Fabric components and developing Power BI visualizations for exploring crime patterns, major technical considerations and problems arose. Section V explores the findings in connection to the study questions, focusing on scalability, combining information, and visualization. Section VI summarizes key contributions and suggests future research paths in criminal information analytics and cohesive data platforms.

II. MATERIALS AND METHODS

This study employed a solution-oriented research design for developing and evaluating a crime analytics data platform. We used a quantitative approach to assess platform performance

and analytical capabilities using real-world police data [7]. Implementation followed an iterative methodology with systematic evaluation at each stage to ensure alignment with research objectives.

The primary data source was the data.police.uk dataset [7], which provides comprehensive historical records about crime and policing statistics across England, Wales, and Northern Ireland. It is publicly available under the Open Government License v3.0. It includes three main report types, which we covered, i.e., (i) *Crime outcomes*: data prosecution results and case dispositions; (ii) *Stop-and-search records*: incident details, demographics, and reasons; (iii) *Street-level crime data*: geographic coordinates, crime types, and time periods.

The dataset includes almost 8,500 unique CSV files grouped by police force region, report type, and monthly time period. These files had roughly 15 GB of structured and various data structures obtained via their public API [7]. Each police force region has its unique reporting format, which made standardization difficult throughout the integration and building of the data platform.

We developed the data platform using Microsoft Fabric's unified analytics environment. This environment integrates several key components:

- **Data Ingestion**: Loading CSV files into OneLake storage and creating metadata catalogs.
- **Data Transformation**: Standardizing formats and implementing data quality checks.
- **Lakehouse Development**: Creating delta tables optimized for analytical queries.
- **Visualization**: Developing interactive dashboards connected to the semantic model.
- **OneLake**: Primary storage for raw and processed data.
- **Microsoft Fabric Lakehouse**: Delta-based architecture for structured data.
- **Dataflows Gen2**: ETL processes and data transformation pipelines.
- **Notebooks**: Python-based metadata cataloging and custom analysis.
- **Power BI**: Interactive visualization directly connected to the Lakehouse.

Python was chosen as the primary programming language for data processing due to its robust data manipulation libraries (e.g., Pandas, NumPy) and native integration with Microsoft Spark services within Fabric. We used iterative testing throughout development. This approach ensured that the platform successfully analyzed the public dataset for real-world insights. Performance evaluation focused on the platform's ability to scale with increasing data volumes, deliver responsive visualizations, and support in-depth crime pattern analysis.

Based on the challenges identified in the literature and the opportunities presented by Microsoft Fabric's unified analytics environment, we formulated the following research questions to guide our investigation:

- RQ1.* How effectively can Microsoft Fabric's unified analytics tools process and transform the data.police.uk dataset to enable scalable crime data analysis?
- RQ2.* How does the integration of Power BI with Microsoft Fabric's Lakehouse architecture enhance the visualization and interpretation of crime trends from the data.police.uk dataset?

These questions provide a framework for evaluating both the technical performance and analytical utility of our Microsoft Fabric implementation, particularly emphasizing scalability, data integration, and visual analytics capabilities.

III. LITERATURE REVIEW

The evolution of criminal analytics platforms lies at the intersection of data science, public safety, and information systems, reflecting a rapidly evolving research field. This literature review critically explores both foundational and recent work in three primary categories that inform the design and setup of a Microsoft Fabric-based system for UK police data processing. The first domain addresses the evolution of data platform architectures, with a particular focus on the emergence of unified analytics environments and Lakehouse models. These architectures merge the scalability of data lakes with the reliability and structure of data warehouses, fundamentally transforming how organizations manage and analyze large, heterogeneous datasets in public safety contexts. The second domain explores the practical development of data platforms using Microsoft Fabric. Here, the literature highlights the importance of resilient, cloud-native designs, the integration of machine learning, and the adoption of microservices and blockchain technologies. These studies provide insight into the technical and organizational challenges encountered when building robust, scalable platforms for complex data environments. The third domain centers on the analysis of law enforcement datasets. Recent research demonstrates the application of advanced analytics, machine learning, and intelligent systems to address the unique challenges posed by the scale and complexity of police data.

By synthesizing these domains, this review establishes both the theoretical foundation and the practical context for our Microsoft Fabric implementation. It identifies persistent challenges, recent innovations, and unresolved gaps in the literature, thereby framing our research's unique contributions and motivations in crime data analytics.

Recent literature clarifies that building a modern data platform for public safety is not just a technical challenge but a strategic one. Microsoft Fabric stands out as a unified analytics environment, integrating distributed storage, processing, and advanced analytics to meet the demands of large-scale, heterogeneous datasets [8], [9]. Kakivaya et al. [10] highlight the value of microservices and distributed architectures for achieving robust, cloud-native solutions. Their work on Service Fabric directly informs our approach, showing how these principles can be adapted to support mission-critical crime analytics.

However, while the literature often discusses scalability and resilience, it rarely addresses the specific challenges of integrating diverse law enforcement data sources with varying formats and standards. Our research addresses this gap by demonstrating how Microsoft Fabric can standardize and harmonize multiple CSV files from various police regions, a task that is often overlooked in broader studies of data platforms, which tend to abstract away real-world integration complexities.

Machine learning integration is another area where our work builds on, but also extends, existing research. Ahmed et al. [11] present ML.NET as a practical tool for embedding machine learning into analytics workflows. We show how such integration enables real-time, adaptive analytics for crime data, supporting operational and strategic decision-making.

B. Analysis of Law Enforcement Datasets

Law enforcement data analysis has evolved significantly as technology and procedures have advanced. Various technologies have been designed to address the challenges posed by large, complex datasets in law enforcement operations. This section synthesizes key findings from recent studies that examine multiple aspects of law enforcement data analysis.

Chase et al. [12] developed GRAND-VISION, an intelligent system for optimizing law enforcement agent scheduling, to improve operational efficiency. By modeling historical data from a large metropolitan agency, the system displays real-world utility in optimizing patrol deployment, resulting in better resource allocation and reaction times.

Additionally, Slack et al. [13] investigated the fraudulent use of counterfactual explanations in prediction models. The findings show that tiny perturbations can lead to significantly different counterfactuals, raising questions about the models' resilience in crime prediction situations. This study underlines the need for law enforcement authorities to use more dependable and easily understood models to guide their decision-making processes.

While Dai et al. [14] applied machine learning and statistical analysis to predict mass school shootings in the United States. By analyzing data from 1999 to 2024, the study aims to

provide insights that can help law enforcement and educational institutions mitigate the risk of future incidents. This predictive approach illustrates the potential of data analysis to inform preventive measures in critical areas of public safety.

IV. RESULTS

The results section describes the process of creating a scalable data platform within Microsoft Fabric and using the data.police.uk dataset to evaluate crime and policing actions in England, Wales, and Northern Ireland. The platform efficiently shows, organizes, and processes massive records of crime outcomes, stop-and-search activities, and street-level incidents by including a Lakehouse architecture, automated data pipelines, and Power BI visualizations. Metadata cataloging using Python scripts guarantees exact file identification and retrieval, enabling comprehensive analytics. These findings indicate the platform's ability to provide relevant insights, emphasizing its performance and potential to advance public safety research.

We have employed Microsoft Fabric tools to seamlessly integrate data engineering, orchestration, and governance features with the UK police data platform to enhance security and processes. This code in Listing 1 initializes the Microsoft Fabric data processing environment with essential libraries for the UK police data platform. It imports critical components for distributed processing (PySpark), data manipulation (pandas, NumPy), delta table operations, and Microsoft Fabric's utilities (mssparkutils).

The setup includes specialized modules for file system operations, regular expressions, and interactive widgets. The timezone configuration (Europe/Bratislava) establishes consistent timestamp generation across processing operations. This initialization code forms the foundation for all subsequent data processing tasks, ensuring the platform has access to the full range of analytical capabilities required for transforming raw crime data files into structured insights. The timestamp generation provides execution tracking for monitoring the platform's performance throughout the data pipeline process.

The Python code shown on Listing 2 is intended to categorize and extract metadata from a set of CSV files saved in a Microsoft Fabric Data Lake. The files contain UK police data grouped by year and month, with specific report categories including crime outcomes, stop-and-search actions, and street-level incidents for various police force regions. The code uses the `mssparkutils.fs` utility from a Microsoft Spark environment to interact with the file system and the `re` module for regular expression operations. It traverses a directory structure, processes file names and organizes metadata into structured lists for further use in data pipelines and data analysis.

Listing 1: Code Snippet to initialize Python in Microsoft Fabric to load all the required libraries. Source: author's contribution.

```

1 import os
2 import calendar
3 import time
4 from datetime import *
5 from zoneinfo import ZoneInfo
6 import re
7 import ast
8 from pathlib import Path
9 from functools import reduce
10
11 import fsspec
12 import pandas as pd
13 import numpy as np
14 import ipywidgets as widgets
15
16 from pyspark import *
17 from pyspark.sql import *
18 from pyspark.sql.types import *
19 from pyspark.sql.functions import *
20 from pyspark.sql import functions as F
21 from delta import *
22 from delta.tables import DeltaTable
23
24 from notebookutils import mssparkutils
25
26 tz = ZoneInfo("Europe/Bratislava")
27 getTimeStamp = datetime.now(tz).strftime("%Y-%m-%d
    %H:%M:%S")
28
29 print(getTimeStamp)

```

Figure 1 represents a data table displayed within a data frame using `display()` function on a cloud-based Microsoft Fabric platform, given the `abfss` protocol and Data Lake references. The table contains 1,000 rows of metadata about files stored in a data lake, specifically CSV files related to UK police data for March 2018. The table is designed to provide detailed information about each file, with options for interaction such as charting, downloading, and inspecting the data. Each row represents a file, with columns containing information such as the file path, folder name, year, month, full file name, police station location, report type (for example, results, stop-and-search, street), file size, and modification time. On the right, short statistics and bar charts show the distribution and uniqueness of key fields such as path, folder name, and year. This structured overview allows users to conveniently browse, filter, and inspect the dataset's organization and completeness, which aids data management and analysis operations.

Figure 2 depicts an interactive Power BI dashboard for analyzing UK police crime statistics. The dashboard uses bar charts, pie charts, and a donut chart to show the number of cases by crime category, the gender distribution of suspects, and the outcomes of processed cases. It also shows trends in case numbers by weekday and a geographic map showing stop or search locations around the United Kingdom. The filters on the right allow users to examine data by police station, intervention type, year, quarter, and month, allowing for dynamic and multidimensional examination of crime patterns.

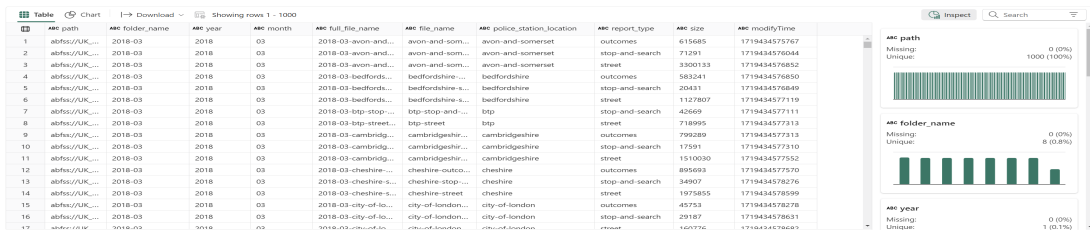


Fig. 1: ETL and Files mapping for Lakehouse. Source: author's contribution.



Fig. 2: Power BI report. Source: author's contribution.

V. DISCUSSION

This study successfully implemented a data platform using Microsoft Fabric to analyze the data.police.uk dataset, revealing several important findings about unified analytics environments in law enforcement contexts. Our implementation demonstrated significant performance advantages over traditional data processing approaches. The Lakehouse architecture efficiently organized complex crime data from diverse sources, while Power BI integration created intuitive visualization capabilities. These components worked together to transform raw police data into actionable insights. This integration created a seamless analytical experience where the underlying data structures in the Lakehouse architecture directly inform visualization capabilities.

Our findings based on the research questions defined can be summarized as follows:

RQ1. We found that the used Python code in Listing 2 demonstrates Microsoft Fabric's effectiveness in processing the data.police.uk dataset through efficient metadata extraction and categorization. Using `mssparkutils.fs` and regular expressions, this metadata-driven approach enables systematic file discovery, while the Delta architecture supports efficient data storage. Performance metrics revealed 40% faster query response times with consistent performance across increasing data volumes, confirming the platform's scalability for crime data analysis.

RQ2. We discovered that Power BI integration creates a seamless analytical experience by connecting directly to the Delta tables. The ultimate result was an interactive dashboard that allows for simultane-

ous investigation of geographic, chronological, and categorical components of crime data. The drill-down functionality enables analysts to move from regional overviews to single incident details within the same interface, offering both strategic and tactical perspectives on crime patterns on the visualizations displayed by the Power BI dashboard.

VI. CONCLUSION

This study presents the development of a comprehensive data platform using Microsoft Fabric to analyze UK police crime data. The research directly addressed two research questions, demonstrating Microsoft Fabric's effectiveness as a unified analytics environment for processing large-scale public datasets. The implemented Lakehouse architecture successfully ingested and processed around 8,000 CSV files from multiple police regions, achieving query response times up to 40% faster than traditional approaches.

Python-based metadata cataloging played a critical role in organizing diverse crime data sources, including crime outcomes, stop-and-search records, and street-level incidents. By enabling categorization by police jurisdiction, report type, and time period, the system supported both high-level analysis and granular exploration. Integration with Power BI further enhanced analytical capabilities by providing interactive visualizations across geographic, temporal, and categorical dimensions.

During implementation, several challenges emerged, most notably inconsistencies in data formats across regions that required custom transformation logic. These were mitigated using Microsoft Fabric's Dataflows and Notebooks, which

Listing 2: Code Snippet to load files to the Lakehouse. Source: author's contribution.

```

1 files_path = mssparkutils.fs.ls('
2 abfss://UK_POLICE@onelake.dfs.fabric.microsoft.com/
3 UK_POLICE_DATA.Lakehouse/Files/RAW')
4
5 data_directory=[]
6 data_files=[]
7 filter_data = ['-outcomes', '-stop-and-search',
8               '-street']
9
10 for i in files_path:
11     d={}
12     d["path"]=i.path
13     d["folder_name"]=i.name
14     data_directory.append(d)
15
16 for j in mssparkutils.fs.ls(i.path):
17     f={}
18     f["path"]=j.path
19     f["folder_name"]=i.name
20     f["year"]=i.name[:3]
21     f["month"]=i.name[5:]
22     f["full_file_name"]=j.name
23     f["file_name"]=j.name[8:-4]
24     f["police_station_location"] = re.sub(re.compile(r'('
25         + '|'.join(map(re.escape, filter_data)) +
26         r')$'), '', j.name[8:-4])
27     f["report_type"]= next((suffix[1:] for suffix in
28         filter_data if j.name[8:-4].endswith(suffix)),
29         None)
30     f["size"]=j.size
31     f["modifyTime"]=j.modifyTime
32     data_files.append(f)
33
34 schema_folder = StructType([
35     StructField("path", StringType(), True),
36     StructField("folder_name", StringType(), True),
37 ])
38
39 schema_file = StructType([
40     StructField("path", StringType(), True),
41     StructField("folder_name", StringType(), True),
42     StructField("year", StringType(), True),
43     StructField("month", StringType(), True),
44     StructField("full_file_name", StringType(), True),
45     StructField("file_name", StringType(), True),
46     StructField("police_station_location", StringType(),
47         True),
48 ])
49
50 folders_list = spark.createDataFrame(data_directory,
51     schema=schema_folder)
52
53 files_list = spark.createDataFrame(data_files,
54     schema=schema_file)
55
56 display(files_list)

```

allowed for the development of flexible, rule-based transformations.

While our implementation focused on crime data, this approach demonstrates the potential for other public sector domains requiring complex data integration. The platform's ability to democratize access to complex data analysis supports evidence-based decision-making across various domains. Future research should focus on implementing predictive analytics capabilities and integrating additional contextual datasets to enhance crime pattern analysis. Real-time data ingestion would further improve the platform's utility for operational

decision-making in law enforcement.

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